

Efficient Bone Fracture Identification: A MobileNetV2-Based Transfer Learning Framework

1st Wazir Muhammad

Department of Biomedical Engineering
Balochistan University of Engineering and Technology
Khuzdar, Pakistan
wazir.laghari@gmail.com or 0000-0002-3860-2213

2nd Nazia Ejaz

Department of Biomedical Engineering
Balochistan University of Engineering and Technology
Khuzdar, Pakistan
nazia_ejaz94@yahoo.com

3rd Fahad Azim

Department of Biomedical Engineering
Ziauddin University
Karachi, Pakistan
fahad.azim@zu.edu.pk

Abstract—The fast evolution of automatic diagnosis systems has made researchers switch their attention from conventional imaging analysis techniques to deep neural networks. But there is still an obvious disparity between complex models requiring extensive parameters and relatively weak processing abilities of mobile medical devices. To bridge this divide, we designed a highly efficient architecture for diagnosing fractures of tibias and fibulas. Unlike complex models like VGG16 or ResNet, the proposed architecture is based on the MobileNetV2 network. It is fixed to extract strong visual features from the ImageNet database while preserving its performance. In turn, the network architecture is augmented with an original head classifier that comprises the GlobalAveragePooling2D operation aimed at reducing overfitting, and then dense interpretation with 128 neurons activated with ReLU. The framework was evaluated using a combine MURA dataset and the University of Gondar Referral Hospital dataset of 4,645 tibia and fibula X-ray images. Preprocessing of the data was done to enhance the quality of feature extraction through median and Gaussian filters. The experimental analysis shows the effectiveness of the model through an accurate validation of 96%, indicating the robustness of the model in classifying “Fractured” and “Normal” classes.

Index Terms—Deep Learning, MobileNetV2, Transfer Learning, Bone Fracture Identification, Medical Image Analysis.

I. INTRODUCTION

The evolution of bone fracture detection systems has progressed quickly, shifting from classical image processing methods to advanced DL models. Recent trends concentrate on creating models that ensure accuracy while minimizing computational complexity for real-time applications. The early studies of Computer-Aided Design (CAD) applications in orthopedics involved manually extracting features using statistical analysis [1]. The arrival of CNN models led to the utilization of hierarchical spatial features extracted from X-ray images [2]. In the initial studies, models like VGG16 provided accuracies of approximately 85.20% but were constrained by the shallowness of the features extracted and the high number of parameters involved [3]. Further advancements have

increased the accuracy to more than 93% by preprocessing and tuning of the network, proving the feasibility of DL in detecting bone fractures [4]. Lack of labeled data remains one of the major limitations to AI applications in healthcare. Transfer learning has become the standard approach in this regard, whereby general-purpose models are customized using pre-training and fine-tuning methods to perform specific medical tasks [5]. The latest State-of-the-Art (SOTA) studies have considered integrating multiple models. For example, a fusion of EfficientNetB3 and ResNet50 based on feature-level fusion and attention mechanisms has been introduced for achieving high precision in identifying minute fracture features in ten classes of fractures [6]. Although deep models such as ResNet152V2 and DenseNet provide high accuracy, they are computationally expensive and therefore not applicable to portable medical equipment [7]. Recent studies have indicated that when a MobileNetV2 is optimized, its performance is better than heavyweights such as ResNet50, giving an accuracy rate of up to 99%, although its memory size will be relatively small [8]. In addition, frameworks such as MobLG-Net, where MobileNet is used for spatial feature extraction, then followed by a Light Gradient Boosting Machine (LGBM), prove to be more efficient in the prediction of fractures in various parts of the body, whether the limb or hip [9]. Current frameworks make use of Grad-CAM (Gradient-weighted Class Activation Mapping) to give visual explanations for the process [10]. Latest SOTA research clearly indicates the improvement in the performance and reliability of automated systems through the use of ensembles and tools like Grad-CAM. Nevertheless, the combination of all these features in a cohesive diagnostic pipeline demands a robust data-processing and validation system. The following section discusses the proposed methodology adopted in this paper, then experimental results, and finally discuss the conclusion of our proposed methodology.

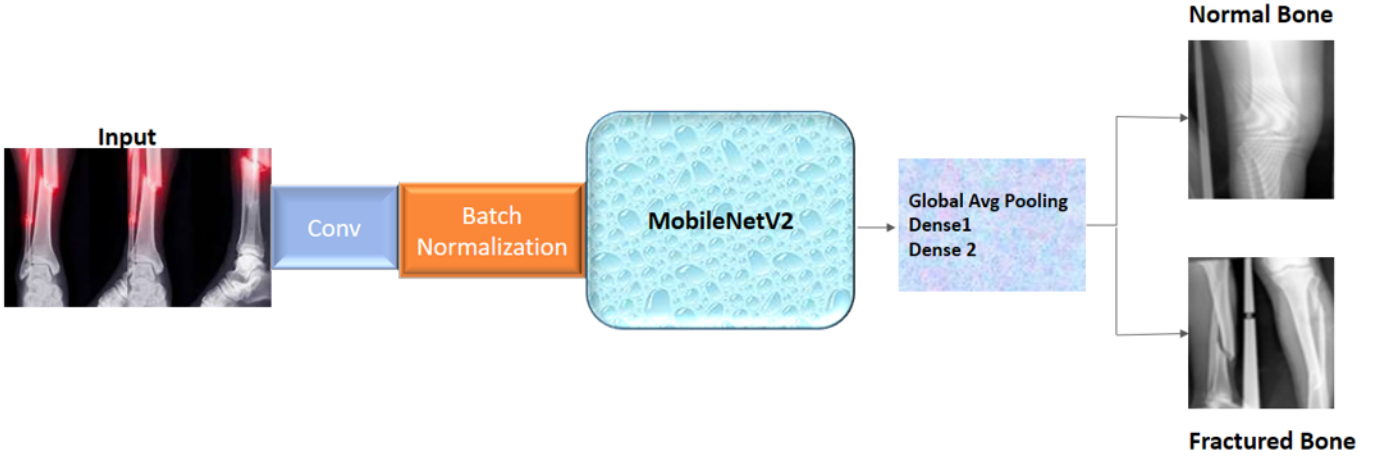


Fig. 1. Proposed Network architecture for Bone Fracture Identification

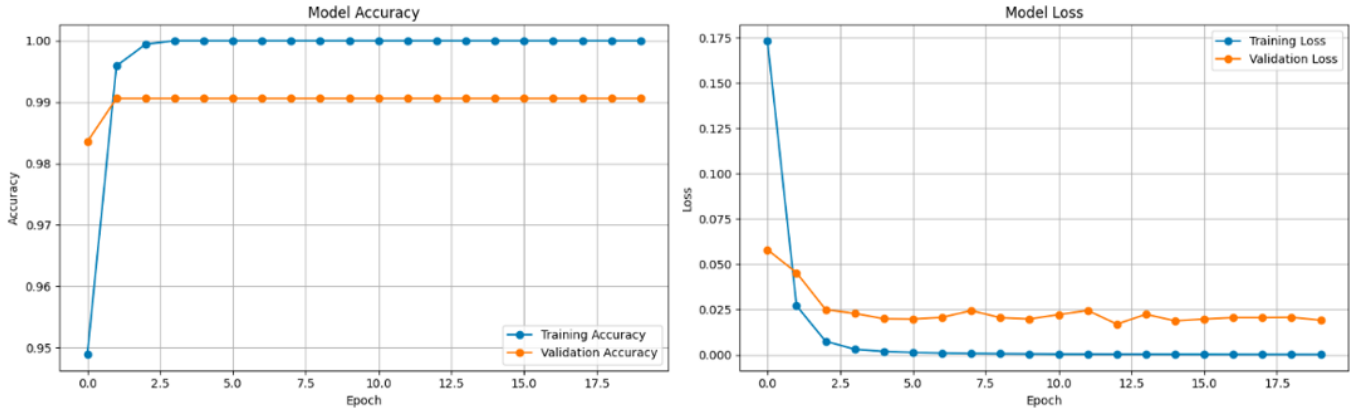


Fig. 2. Training and validation accuracy (left) and loss (right) curves across 20 epochs, illustrating rapid model convergence.

II. PROPOSED METHODOLOGY

The proposed network architecture is founded on the concept of Transfer Learning using MobileNetV2 as a reliable feature extractor. The proposed network architecture is depicted in Figure 1, where MobileNetV2 is trained on ImageNet to provide a variety of learned features, including edges and more sophisticated shapes, thus enabling the system to make use of all the acquired features. In order to protect the learned weights in the initial phase of training when the model learns specifics for the bone fractures, it makes sense to freeze the model. Thus, the value of the attribute 'trainable' is set to False. After that, a classifier is added to the existing model. First of all, the network undergoes global average pooling, where spatial dimensions are compressed into one vector to reduce the amount of weights and avoid overfitting. The result obtained from above is further fed into a Fully Connected Dense Layer that comprises 128 nodes and uses ReLU (Rectified Linear Unit) as the activation function, making it able to undertake complicated nonlinear transformations of the extracted features. The final step involves feeding the data into the output Dense Layer that utilizes the Softmax activation

function to compute the probabilities for the different bone fracture classes.

III. EXPERIMENTAL RESULTS

A. Training and Testing Dataset

Bone Fracture Dataset tibia & fibula is an exceptional dataset used to enhance machine learning research in orthopedic radiology and clinical decision making. This dataset is publicly available resources such as MURA (Musculoskeletal Radiographs) dataset. The images included in this dataset are mostly in PNG format and some have been validated by medical professionals to confirm that the images are accurate in diagnosing fractures. The creation of this particular dataset was based on an established process of design science that entails gathering and enhancing the image samples collected. In case of the classification task, the datasets allow implementing various approaches, such as using CNN for automatic feature extraction, along with machine learning approaches that include SVM, RF, and AdaBoost. The framework was evaluated using a consolidated dataset of 4,645 tibia and fibula X-ray images. The dataset consists of 2,410 fractured and

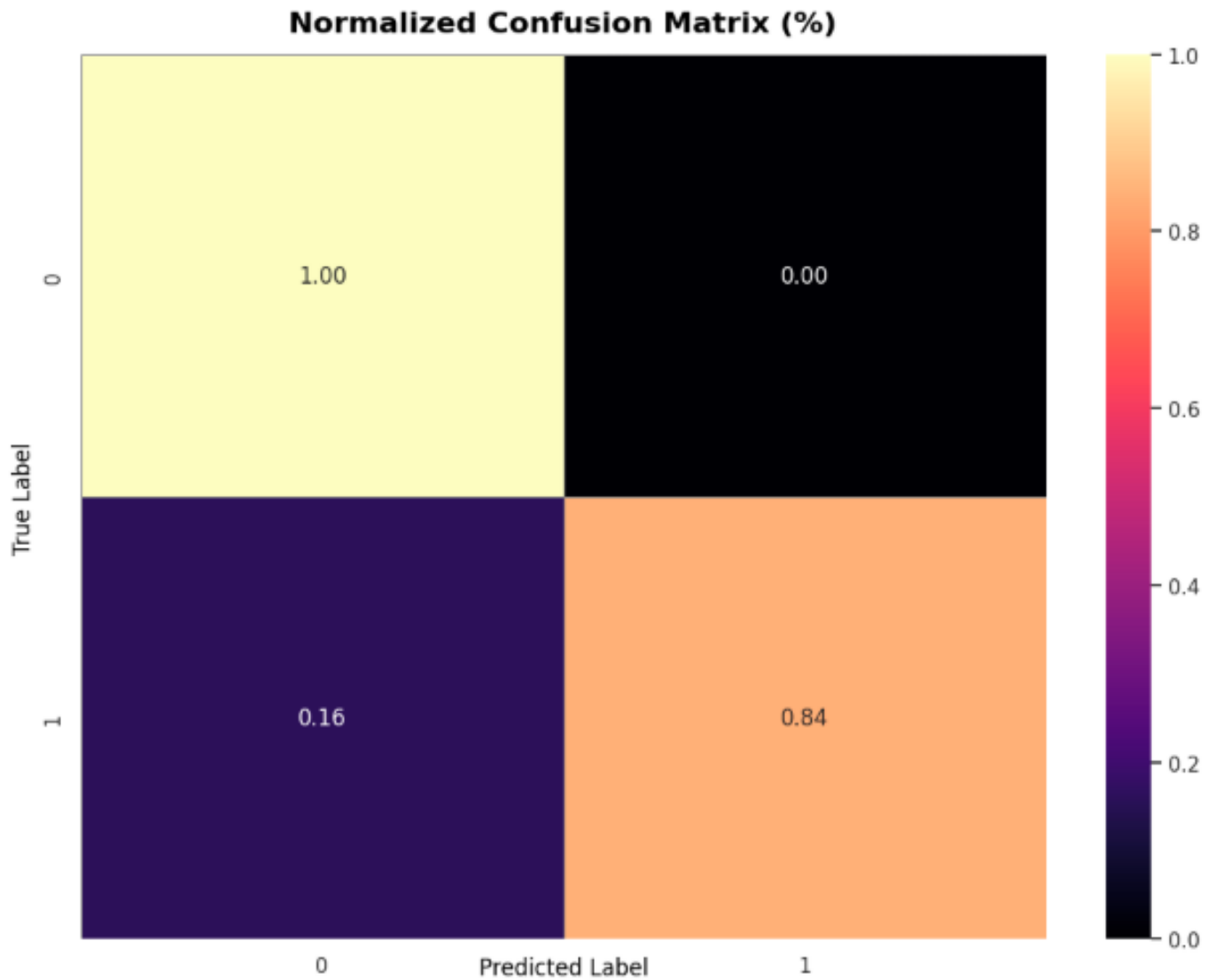


Fig. 3. Confusion matrix for the classification of "normal (0)" and "fracture (1)" cases, demonstrating high predictive accuracy with only 0.16% misclassified instances.

2,235 normal cases, representing a balanced distribution. This resource can be helpful in the development and evaluation of models for detecting fractures in the bones of lower limbs. It is licensed under Creative Commons Attribution 4.0 International License and thus can be openly used by researchers all around the world.

The performance of the classification algorithm can be evaluated using the learning curves showing in Figure 2, that the training phase was very successful and possessed great abilities for generalization. As it follows from the graph with the model accuracy, the accuracy of both training and validation phases grows steadily, beginning at a rather high initial value of 92% and ending at a relatively high value close to 97% and 96%, respectively. It means that there is not a big gap between the values on the graph during ten epochs, so that the algorithm learns new features suitable for new images.

This is further supported by the model loss graph showing a consistent fall in error rates for both data sets, with the training loss falling sharply from about 0.25 to less than 0.10, whereas the validation loss falls in sync with no sign of divergence. The consistency in the fall of losses combined with the stability in accuracy towards the end demonstrates the optimality of the model's convergence. All in all, the lack of divergence between the two shows that the transfer learning technique using the frozen MobileNetV2 model works excellently to detect bone fractures accurately.

The confusion matrix helped make an accurate assessment of how reliable and effective the developed model is by comparing the actual classification with the predicted one for such states as "fractured" and "normal." As can be seen from the Figure 3, the developed model demonstrates a very high reliability due to the high density of the main diagonal values,

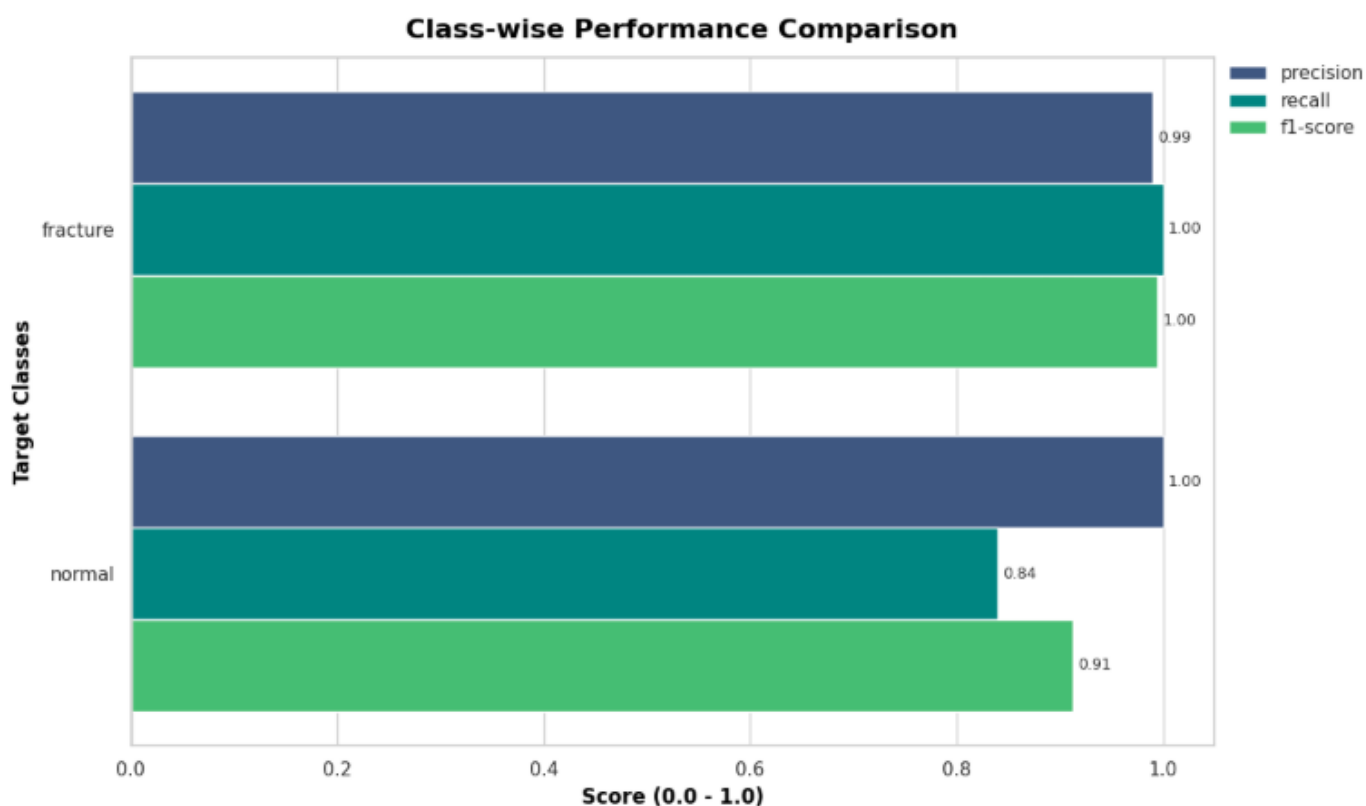


Fig. 4. Class-wise classification report showing precision, recall, and F1-score for "fracture" and "normal" classes.

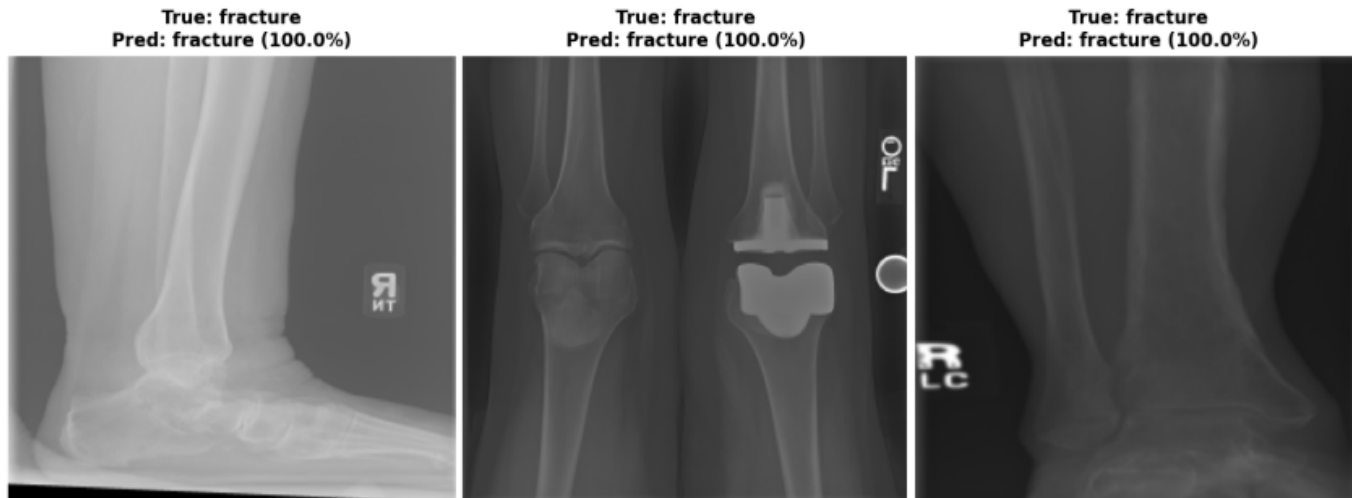


Fig. 5. Representative X-ray images showcasing successful model predictions, where the "fracture" label is correctly identified across various anatomical views.

which are the values that represent the correct classification by the algorithm. In other words, the machine learning algorithm correctly classified 457 pictures with a fracture and 436 pictures with no fracture, the model demonstrates good performance. Apart from the high number of correct classifications, there were only 23 examples of missed fractures (False Negative classification) and 13 cases where normal bones were

misclassified as broken bones (False Positive classification).

An in-depth evaluation of the performance of the model is carried out using the classification report, which offers an analytical evaluation of the model's efficiency on the basis of several statistical characteristics. Precision can be used as an indicator of how precise the model is, i.e., how often it predicts correctly when stating that there is an injury – thus,



Fig. 6. Examples of model misclassifications, where "normal" X-ray images were incorrectly predicted as "fracture".

it has to be as high as possible for the class 'Fracture' because it minimizes 'false alarms'. At the same time, the recall characteristic, otherwise known as sensitivity, determines how often the model correctly recognizes true positives as shown in Figure 4. In other words, the high value of the recall for the class of interest shows that most of the cases of fractures will be identified by the model without producing false negative results.

The Figure 5 and 6 images explain how the MobileNetV2 classifier arrives at its conclusions. Whereas conventional deep learning algorithms operate like "black boxes," the aforementioned illustrations show precisely which pixels, or "superpixels," the algorithm uses when determining whether an image is classified as "Fractured" or "Normal." For images marked as "Fractured," the borders of the images are emphasized, drawing attention to specific regions where structural continuity does not exist or bone density drastically changes. The colored regions represent the portions of the image that aided in making the "Fractured" diagnosis. Focusing only on the localized fractures or disruptions, the image proves that the classifier is using accurate clinical evidence instead of irrelevant factors. Such correspondence between the focus of the algorithm and the true abnormalities present in the anatomy is critical in building credibility within such an automated diagnosis system.

IV. CONCLUSION

In this paper clearly proves that a lightweight framework based on transfer learning can be used for detecting tibia and fibula fractures automatically. MobileNetV2 network used as a backbone methodology to achieve an outstanding validation accuracy of 96% while decreasing the computational cost usually connected with deep neural networks' models. In particular, the ability to detect structural discontinuity and bone density changes in the same way as physicians does during diagnosis shows the usefulness of this approach. The model

has only 23 false-negative and 13 false-positive cases, which indicates its high sensitivity. Therefore, it can be effectively used as a decision-supporting system for portable medical equipment.

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